# Assignment-1

1. In a mobile antenna the magnetic field of the signal is given by

$$H_x = 0.25 \sin\left[\pi(ct - y)\right] A/m$$

$H_y = 0 = H_z$

Where c is the velocity of light in free space. Find the displacement current density. If the antenna is ~~point~~ a circle of radius 2m and ~~its surface~~ the outward normal of the antenna's surface is +ve z direction, then find the displacement current at time $t = 10^{-8}$ s and $y = \frac{5}{2}$ m.

2. In a large power distribution transformer the magnetic field at a point in the air space is $\vec{B} = 0.8 \cos\left[\frac{\pi}{4} \times 10^{-6}(ct - x)\right]$ T. Find the amplitude of the displacement current density.

3. In a large, oil filled capacitor the electric field ~~inside~~ at a point inside the capacitor is given by

$$\vec{E} = 0.6 \sin\left[\frac{\pi}{4} \times 10^{-6}(ct - 5z)\right]\hat{z}$$

V/m. Find the displacement current density and its magnitude.

An electric field is given by

$$\vec{E}(x,y,z,t) = -0.5\cos(kz-\omega t)(\hat{i}+\hat{j})$$

V/m, in the source-free case for which $\rho(x,y,z,t)=0$, $\vec{J}(x,y,z,t)=0$ for all $(x,y,z,t)$, where $\omega$ is the frequency of the electric field. Let $\Gamma$ be the circular boundary of a disc of radius $a$ in the $x$-$y$ plane with centre at the origin and counter-clockwise around the $z$-axis. If $\vec{B}(x,y,z,t)$ is the induced magnetic field corresponding to the above electric field, then derive the circulation of the magnetic field around $\Gamma$ as a function of time $t$.

[Hint: you are required to evaluate the line integral $\oint_\Gamma \vec{B}\cdot d\vec{l} = ??$]

5. The most general form of a plane wave ~~equation~~ in 3-D is given by

$$\vec{E}(\vec{r},t) = \vec{E}_0 \, f(\vec{K}\cdot\vec{r} - \omega t) \quad --- \text{(A)}$$

where $\vec{K} = K_x\hat{x} + K_y\hat{y} + K_z\hat{z}$ is the wave vector, $\vec{E}_0 = E_0(\hat{x}+\hat{y}+\hat{z})$ is a constant vector, ~~and~~ $f$ is an arbitrary function, and $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$ (where $\hat{x}$, $\hat{y}$ and $\hat{z}$ are unit vectors along $x$-$y$-$z$ axis respectively).

(a) Show that the plane wave given by equation (A) satisfies the wave equation $\nabla^2\vec{E} = \frac{1}{c^2}\frac{\partial^2\vec{E}}{\partial t^2}$ as long as the dispersion relation $\omega^2 = c^2k^2$ is satisfied, where $k^2 = (K_x^2 + K_y^2 + K_z^2)$.

(b) Although the wave equation can be derived from Maxwell's equations (proved in your class), this does not guarantee that every plane wave solutions of the wave equation is a solution of Maxwell's equations. In general, other conditions are also need to be satisfied. What other condition/conditions ~~must~~ the above plane wave in Eq. (A) must satisfy in order to be also a solution of Maxwell's equations?

(c) Show that the electric field $\vec{E} = 0.2\cos(kx - \omega t)\,\hat{x}$ V/m satisfies the wave equation but not Maxwell's equations.

6. Moving Loop in time-varying field:

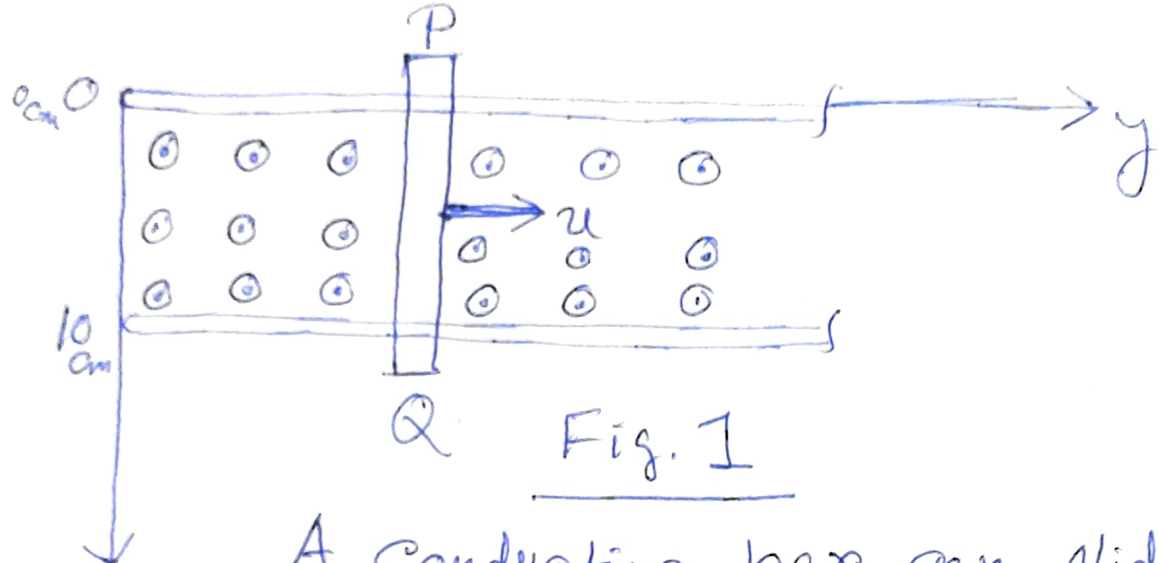


Fig. 1

A conducting bar can slide freely over two conducting rails as shown in fig. 1. Now your task is to calculate induced voltage in the conducting bar for the following cases

(i) If the bar is stationed at $y = 20$ cm and $\vec{B} = 5\cos(\omega t)\,\hat{z}$ mWb/m$^2$ with $\omega = 10^6$ Hz and $t = 10^{-6}$ s.

(ii) If the bar slides at a velocity $u = 10\,\hat{y}$ m and $\vec{B} = 6\,\hat{z}$ mWb/m$^2$

(iii) If the bar slides at a velocity $u = 20\,\hat{y}$ m/s and $\vec{B} = 6\cos[(\omega t - ky)\pi]\,\hat{z}$ mW at $t = 0.01$ s, $\omega = 10^2$ Hz and $k = 2.5$/m